

Mohamad Chehade

Research Curriculum Vitae

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Research focus: trustworthy AI for consequential decisions. I develop methods to verify safety, adapt behavior at inference time, and evaluate models by downstream decision quality.

Education

The University of Texas at Austin

Ph.D., Electrical and Computer Engineering

Master of Science in Engineering, Aug. 2023 – May 2026. Advisor: Dr. Hao Zhu.

Austin, TX

Aug. 2023 – Present

American University of Beirut

B.Eng., Electrical and Computer Engineering

GPA: 4.0/4.0; Minor in Mathematics. Final-year project: Optimal Power Flow via Machine Learning (Dr. Rabih Jabr).

Undergraduate research: Microgrid Sizing using Ordinal Optimization (Dr. Sami Karaki).

Beirut, Lebanon

Aug. 2019 – Jun. 2023

Skills and Relevant Coursework

Programming: Python, C/C++, MATLAB, SQL, R

ML and scientific computing: PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Matplotlib

Optimization: Gurobi; convex and mixed-integer optimization, ordinal optimization

Power systems: PowerWorld, MATPOWER, Simulink, SPICE, HOMER, PVSyst, LabVIEW

Development: Git, Linux, LaTeX, Overleaf, Microsoft Office Suite

Languages: English (fluent), French (intermediate), Arabic (native)

Coursework: Reinforcement Learning; Generative AI; Statistical Machine Learning; Applied Machine Learning; Probability and Stochastic Processes; Applied Stochastic Processes; Learning-based Optimal Control; Convex Optimization; Energy Optimization and Operation

Research and Engineering Experience

The University of Texas at Austin

Graduate Research Assistant – Advisor: Dr. Hao Zhu

Austin, TX

Aug. 2023 – Present

- **Risk adaptation (TRAM):** Developed test-time policy composition to adjust reward–risk trade-offs without re-training source agents.
- **LLM alignment (SITAlign):** Developed inference-time alignment that optimizes a primary objective subject to thresholds on other criteria.
- **LLMs for grid actions:** Fine-tuned language models with supervision and preferences to propose feasible corrective switching plans.
- **Distribution-grid control (NEO-Grid):** Built neural power-flow surrogates and deep equilibrium models for voltage optimization and control.
- **Battery control:** Compared model-free and model-based controllers for battery operation under time-varying electricity prices.
- **Carbon-aware energy management:** Co-developed market-participation methods that weigh building energy costs against carbon emissions.

Argonne National Laboratory

Research Aide – Grid Optimization and Infrastructure Group

Mentors: Dr. Feng Qiu and Dr. Wei Gao.

Remote

May – Aug. 2026

- Developed decision-aware time-series foundation models to align forecasts with downstream decision-making and optimization.

Los Alamos National Laboratory

Graduate Research Assistant – T-5 Applied Mathematics and Plasma Physics

Mentor: Dr. Russell Bent.

Remote

May – Aug. 2025

- Developed robustness-training methods that use verifiable input regions to improve neural-network certified robustness.

Los Alamos National Laboratory

Graduate Research Assistant – T-5 Applied Mathematics and Plasma Physics

Mentors: Dr. Wenting Li and Dr. Brian Bell.

Los Alamos, NM

Jun. – Aug. 2024

- Developed two algorithms for large exact verifiable input regions of neural networks, targeting physical and safety-critical systems (LEVIS, ICML 2025).

- Developed inverter fault diagnosis using combinational logic with industrial motor manufacturer Ward Leonard.
- Optimized and built an inverter circuit for high-power testing using mixed-integer linear programming.

Publications – Published or Accepted

1. **M. Chehade**, A. S. Bedi, S. Chakraborty, A. Zhang, and H. Zhu. Test-Time Risk Adaptation with Mixture of Agents. *NeurIPS 2026 (accepted)*.
2. **M. Chehade**, W. Li, B. W. Bell, R. Bent, S. R. Kazi, and H. Zhu. LEVIS: Large Exact Verifiable Input Spaces for Neural Networks. *ICML 2025*.
3. **M. Chehade**, S. S. Ghosal, S. Chakraborty, A. Reddy, D. Manocha, H. Zhu, and A. S. Bedi. Bounded Rationality for LLMs: Satisficing Alignment at Inference-Time. *ICML 2025*.
4. **M. Chehade** and H. Zhu. Fine-Tuning LLMs to Generate Economical and Reliable Actions for the Power Grid. *IEEE PES General Meeting 2026 (accepted)*.
5. **M. Chehade** and H. Zhu. NEO-Grid: A Neural Approximation Framework for Optimization and Control in Distribution Grids. *Hawaii International Conference on System Sciences (HICSS), 2026*.
6. Y.-H. Cho, **M. Chehade**, F. Al-Janahi, S. Lim, J. Mohammadi, and H. Zhu. Carbon-aware Market Participation for Building Energy Management Systems. *IEEE PES General Meeting 2026 (accepted)*.
7. **M. Chehade** and S. Karaki. BOOST: Microgrid Sizing Using Ordinal Optimization. *IEEE Texas Power and Energy Conference (TPEC), 2025, pp. 1–4*.
8. **M. Chehade**, Y.-H. Cho, S. Chinchali, and H. Zhu. Should We Use Model-Free or Model-Based Control? A Case Study of Battery Control. *North American Power Symposium (NAPS), 2024, pp. 1–5*.

Awards and Honors

2026 Third place, American Society of Mechanical Engineers (ASME) CIE Student Hackathon – AI for engineering design and manufacturing.
2024 Best Graduate Presentation Award, North American Power Symposium (NAPS).
2023 Mohamad Ali Safieddine Award for Academic Excellence – ranked first across the American University of Beirut Faculty of Engineering and Architecture.
2023–present Cockrell School of Engineering Fellowship, UT Austin.
2023 Electrical and Computer Engineering Distinguished Graduate Award – ranked first among ECE graduates.
2023 Exceptional ECE Final Year Project Award – Power and Energy Systems.
2021 Best Pitch Award, EPFL Tech4Impact Summer School (Swiss Federal Institute of Technology Lausanne).
2019–2023 Dean's Honor List, American University of Beirut – every awarded semester.

Tutorial

Upcoming, October 26, 2026 – IEEE SmartGridComm, College Station, TX. Agentic AI for Grid Operators in Active Distribution Networks within a Multi-Agent Framework. Co-presenters: Dr. Alaa Selim (MIT) and Dr. Mahmoud Abdelaal (ABB).

Peer Review Service

Neural Information Processing Systems (NeurIPS)	June 2026 – Present
International Conference on Machine Learning (ICML)	Feb. 2026 – Present
International Conference on Learning Representations (ICLR)	Oct. 2025 – Present
AAAI Conference on Artificial Intelligence	Aug. 2026 – Present
IEEE Transactions on Smart Grid	Sep. 2023 – Present
IEEE Power Engineering Letters	May 2026 – Present
IEEE Systems Journal	Sep. 2025 – Present
Hawaii International Conference on System Sciences (HICSS)	June 2025 – Present
Asilomar Conference on Signals, Systems, and Computers	July 2024 – Present
IEEE Texas Power and Energy Conference (TPEC)	Dec. 2024 – Present